

HENGYI YIN

646-246-7508 | hengyi.yin@columbia.edu | Personal Website | LinkedIn | GitHub

EDUCATION

Columbia University

Master of Science in Electrical Engineering

New York, US

Aug. 2026 – Dec. 2027

- Selected graduate coursework: Reinforcement Learning; Heterogeneous Computing for Signal and Data Processing; LLM-Based Generative AI; Systems and Networks for Large-Scale LLM Inference.

McGill University

Bachelor of Engineering in Electrical Engineering, Minor in Economics

Montreal, CA

Sep. 2021 – Jun. 2026

RESEARCH EXPERIENCE

The Neuro (Montreal Neurological Institute-Hospital)

Machine Learning Research Assistant

Jan. 2025 – Dec. 2025

Montreal, CA

- Worked on quantitative MRI and quantitative susceptibility mapping (QSM) pipelines, including image preprocessing, brain masking, image registration, and downstream quantitative analysis.
- Investigated brain-mask generation for QSM and evaluated **BEN**, a deep learning-based brain extraction method, as an alternative to conventional masking approaches.
- Developed and evaluated neuroimaging preprocessing workflows using Python and imaging tools including FSL, ANTs, NiBabel, and SimpleITK, with an emphasis on robust and reproducible analysis.

Central South University, School of Traffic & Transportation Engineering

Research Assistant

Apr. 2023 – Sep. 2023

Hunan, CN

- Developed data-driven estimation methods for DC-link capacitor health monitoring using MATLAB and Simulink.
- Designed optimization and simulation procedures for capacitance estimation under low sampling-frequency measurements, contributing to a peer-reviewed publication in *IEEE Transactions on Power Electronics*.

PUBLICATIONS

- X. Wu, K. Li, R. Tian, **H. Yin**, T. Yu, S. Cheng, and C. Chen, “A Capacitance Estimation Method for DC-Link Capacitors in Railways Based on Precharging Model and Low Sampling Frequency,” *IEEE Transactions on Power Electronics*, vol. 39, no. 1, pp. 1527–1537, Jan. 2024.

INDUSTRY EXPERIENCE

Huawei Technologies Co., Ltd.

Machine Learning Engineer Intern

Jun. 2024 – Sep. 2024

Shanghai, CN

- Developed LLM-based knowledge retrieval and code-assistance workflows using Retrieval-Augmented Generation (RAG), embeddings, and semantic search.
- Built data-processing, code-understanding, and error-analysis tools in Python, Java, and Go to support software debugging and technical-information retrieval.

SELECTED PROJECTS

Graph Neural Networks for Spatiotemporal Forecasting | PyTorch

2025

- Implemented and evaluated graph neural network architectures for spatiotemporal forecasting on traffic and mobility data, including training and evaluation pipelines in PyTorch.

LLM Retrieval System | Hugging Face, Sentence Transformers

2024

- Built a Retrieval-Augmented Generation pipeline using dense embeddings and semantic search; achieved 3rd place among 76 teams at McGill CodeJam14.

TECHNICAL SKILLS

Programming: Python, C, Java, Go, SQL, MATLAB **Machine Learning:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, OpenCV, NumPy, Pandas

Biomedical Imaging: FSL, ANTs, NiBabel, SimpleITK, Quantitative MRI, QSM, Brain Extraction, Image Registration

Research Interests: Multimodal Learning, Generative Models, Computer Vision, Robust Machine Learning, Biomedical Imaging

Tools: Git, Docker, AWS, Jupyter, VS Code, MATLAB/Simulink